Executive Function as Unity and Diversity

Executive function is a family of related but not identical control processes (inhibition, working memory, shifting); early performance is context-bound and is not intelligence, obedience, or character.

-CONSTRUCT; MODERATE-TODDLER-STRUCTURE; LOW-SINGLE-TASK | STATUS: DRAFT | DOMAIN: EXECUTIVE-FUNCTION, COGNITIO

WHAT WE KNOW (Narrowest Supported Conclusion)

Executive function (EF) names top-down processes that help maintain goal-directed behavior when a more automatic response is insufficient. Miyake et al. (2000) showed in adults that inhibition, updating, and shifting share variance and are separable. Garon, Bryson & Smith (2008) mapped an integrative, developmental framework for preschoolers: early forms of attention, inhibition, and working memory are visible before a stable adult factor structure. Toddler performance is a precursor, not miniature adult EF.

EVIDENCE AND CAUSAL ASSESSMENT

The evidence is psychometric, experimental, and developmental, not a causal law of parenting. Task performance includes language, speed, motor demands, motivation, and familiarity. Garon et al. (2008) is a review, not a new RCT. Delay-of-gratification prediction is a different, attenuated claim on the Watts card.

EFFECT SIZE AND PERSISTENCE

Age-related gains across early childhood are robust at the population level. Rank-order stability is lower in toddlerhood than later. A developmental increase is not a permanent individual difference.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a for the construct claim. Task impurity, low reliability, cultural familiarity, and language load are the main threats. SES–EF associations exist in other literatures and are not this card's identification.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

EF is not intelligence, obedience, moral worth, or temperament. High compliance can occur without flexible goal maintenance. Poor task performance can reflect stress, language, or an invalid task. Early differences are not a verdict on children or parents.

WHAT ACTION IS REASONABLY JUSTIFIED

Use multiple observations across contexts. High-stakes ranking of toddlers by one EF task does not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- Garon, Bryson & Smith (2008), Psychological Bulletin
- Miyake et al. (2000), Cognitive Psychology